Adaptive Hybrid Control Error for Vehicle Platoons via Reinforcement Learning

Mid-Report

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1 Introduction

1.1 A Short Description

This project studies mixed traffic on a ring road where human-driven vehicles (HDVs) and connected autonomous vehicles (CAVs) share the same lane. The objective is to reduce traffic-wave amplification without giving up the robustness of a standard longitudinal controller.

Our approach keeps a microscopic car-following controller at vehicle level and adds a mesoscopic adaptation layer that reacts to upstream traffic conditions. Reinforcement learning is used only as a residual correction around a rule-based baseline, so the controller remains interpretable while still being able to adapt to traffic patterns that are difficult to encode by hand.

1.2 Motivation

A recurring difficulty in platoon control is the trade-off between robustness and anticipation:

- **Local sensing:** Radar/LiDAR-based control is robust to communication failures and is well suited to collision avoidance, but it often struggles to damp disturbances once they propagate through a long platoon.
- **Global cooperation:** Leader or platoon-wide information can improve string stability, but its benefits deteriorate quickly when communication is delayed, intermittent, or noisy.

1.3 Our contributions

- A mixed-traffic ring-road model combining SIDM-based HDVs, ACC/CTH-based CAVs, and mesoscopic traffic indicators derived from upstream flow.
- A layered control architecture in which a rule-based mesoscopic headway adaptation is refined by a residual RL policy, rather than replaced by a fully learned controller.
- A corrected and admissible evaluation protocol at 50 % human penetration, leading to the first clean comparison in this project between the no-RL baseline, PPO, and SAC.

2 Traffic Environment

2.1 Driver Models

2.1.1 Intelligent Driver Model (IDM)

The Intelligent Driver Model (IDM) is a deterministic microscopic car-following model describing the longitudinal dynamics of vehicle i as a function of its own velocity, the spacing to its predecessor, and their relative velocity [1]. The acceleration of vehicle i is defined as

$$\dot{v}_i = a_{max} \left[\underbrace{1 - \left(\frac{v_i}{v_d}\right)^{\delta}}_{\text{free-road acceleration}} - \underbrace{\left(\frac{s^*(v_i, \Delta v_i)}{s_i}\right)^2}_{\text{interaction (braking) term}} \right]. \tag{1}$$

Here v_i is the ego velocity, v_d the desired free-flow velocity, a_{max} the maximum acceleration, b the desired deceleration, and δ the acceleration exponent [1]. The free-road term drives the

vehicle toward v_d, while the interaction term enforces deceleration when the actual spacing s_i falls below the desired spacing s^*.

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{2}$$

where x_i and x_{i-1} denote the longitudinal positions of vehicle i and its leader, respectively, and L is the vehicle length. The relative velocity is defined as $\Delta v_i = v_i - v_{i-1}$.

The desired dynamic spacing is given by

$$s^*(v_i, \Delta v_i) = \underbrace{s_0}_{\text{jam distance}} + \underbrace{v_i T_H}_{\text{time-headway term}} + \underbrace{\frac{v_i \Delta v_i}{2\sqrt{a_{max} b}}}_{\text{anticipation term}} . \tag{3}$$

The terms represent the jam distance s_0, the time-headway policy $v_i T_H$, and an anticipative braking correction [1].

2.1.2 Stochastic Intelligent Driver Model (SIDM)

To account for perception errors, behavioral variability, and unmodeled disturbances, the IDM can be extended by introducing a stochastic acceleration component [2]. The stochastic IDM (SIDM) is formulated as

$$\dot{v}_i = \underbrace{f(s_i, v_i, v_{i-1})}_{\text{deterministic IDM}} + \underbrace{\xi_i(t)}_{\text{stochastic acceleration noise}} , \tag{4}$$

where $f(\cdot)$ denotes the deterministic IDM acceleration function and $\xi_i(t)$ is a stochastic process representing white acceleration noise.

The stochastic term satisfies

$$\langle \xi_i(t) \rangle = 0, \tag{5}$$

$$\langle \xi_i(t) \xi_j(t') \rangle = Q\, \delta_{ij}\, \delta(t - t'), \tag{6}$$

where Q denotes the noise intensity, δ_{ij} is the Kronecker delta ensuring independence between different vehicles, and $\delta(t-t')$ is the Dirac delta distribution representing temporal white noise [2].

Close to the string stability threshold, noise can generate spatiotemporal correlations and traffic oscillations resembling stop-and-go waves. Consequently, the SIDM provides a unified framework for analyzing deterministic instabilities and noise-induced traffic dynamics [2].

2.1.3 Connected Autonomous Vehicle (CAV) Model

For connected autonomous vehicles, we adopt the point-mass longitudinal dynamics described in [3]. The motion of vehicle i is governed by

$$\dot{x}_i = v_i, \qquad \dot{v}_i = u_i, \tag{7}$$

In a car-following configuration, the spacing and relative velocity are defined as

$$s_i = x_{i-1} - x_i - L, \qquad \Delta v_i = v_i - v_{i-1}. \tag{8}$$

CAVs implement an Adaptive Cruise Control (ACC) strategy based on a Constant Time Headway (CTH) policy. The desired spacing is given by

$$s_i^{\text{ref}} = s_0 + T_C v_i, \tag{9}$$

where s_0 denotes the standstill spacing and $T_C > 0$ is the selected time headway.

Defining the spacing error

$$e_{p,i} = s_i - s_i^{\text{ref}}, \tag{10}$$

its dynamics follow from

$$\dot{s}_i = -\Delta v_i, \qquad \dot{s}_i^{\text{ref}} = T_C \dot{v}_i = T_C u_i, \tag{11}$$

which yields

$$\dot{e}_{p,i} = -\Delta v_i - T_C u_i. \tag{12}$$

A linear ACC controller that regulates both spacing and velocity errors can therefore be written as

$$u_i = k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i, \tag{13}$$

where $k_s > 0$ and $k_v > 0$ denote the spacing and velocity feedback gains, respectively. This controller constitutes a linear realization of the CTH policy consistent with [3]. The CTH strategy guarantees weak string stability ($\|H(j\omega)\|_\infty \leq 1$) using only local measurements [4].

2.2 Mixed Traffic Model

We consider a mixed traffic system composed of N vehicles circulating on a single-lane ring road of length L_{ring}. The vehicle indices satisfy periodic boundary conditions such that vehicle 1 follows vehicle N.

Let $x_i(t)$ and $v_i(t)$ denote the position and velocity of vehicle $i \in \{1, \ldots, N\}$. The ring-road constraint is given by

$$x_{i-1}(t) > x_i(t), \qquad x_0(t) := x_N(t) + L_{\text{ring}}. \tag{14}$$

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{15}$$

where periodicity applies for $i = 1$.

Let $\mathcal{H}$ denote the set of HDVs and $\mathcal{C}$ the set of CAVs, with penetration rate

$$P = \frac{|\mathcal{C}|}{N}. \tag{16}$$

Human-driven vehicles are modeled according to the stochastic Intelligent Driver Model introduced in Subsection 2.1.2.

HDVs do not utilize acceleration information of the preceding vehicle, reflecting the absence of communication capabilities. Furthermore, HDVs are characterized by a larger effective reaction time T_H, representing the delay associated with human perception and actuation. Consequently, their response to traffic perturbations is slower and more susceptible to amplification of stochastic fluctuations.

Connected autonomous vehicles are modeled according to the ACC framework introduced previously in 2.1.3. In contrast to HDVs, CAVs are assumed to exhibit higher sensitivity coefficients and a shorter desired time headway T_C, reflecting their faster sensing [5]. The condition $T_C < T_H$ therefore captures the higher responsiveness of CAVs compared to HDVs.

When a CAV follows an HDV, communication-based acceleration information is unavailable. In this degraded configuration, the feedforward gain is set to $k_a = 0$, and the effective time headway is increased to $T_C^{\text{deg}} > T_C$, representing reduced cooperative capability. This modeling choice captures the performance degradation of CAVs in mixed traffic reported in [5] and [6].

The closed-loop mixed system is therefore

$$\dot{v}_i = \begin{cases} f_{\mathrm{IDM}}(s_i, v_i, v_{i-1}) + \xi_i(t), & i \in \mathcal{H}, \\ k_s(s_i - s_i^{\mathrm{ref}}) - k_v \Delta v_i + k_a a_{i-1}, & i \in \mathcal{C}. \end{cases} \tag{17}$$

2.3 Macroscopic Information

Up to this point, only microscopic vehicle interactions have been considered. To improve the performance of the proposed framework, macroscopic traffic information are introduced according to [3]. In particular, we define the mean velocity on the ring as

$$\bar{v}(t) = \frac{1}{N} \sum_{i=1}^{N} v_i(t), \tag{18}$$

and the velocity variance

$$\sigma_v^2(t) = \frac{1}{N} \sum_{i=1}^{N} \left(v_i(t) - \bar{v}(t)\right)^2. \tag{19}$$

The mean velocity represents the macroscopic traffic state, while the variance quantifies the level of speed dispersion and therefore the intensity of oscillations. As discussed in [3], the variance provides a compact indicator of transient traffic behavior and string instability.

3 Reinforcement Learning

In this project, reinforcement learning (RL) is used to improve the behavior of connected and automated vehicles (CAVs) in mixed traffic. The main idea is not to replace the baseline controller, but to add an adaptive layer on top of it. This learned layer adjusts the time headway policy according to the current traffic situation. In this way, the controller keeps its physical interpretation while gaining the ability to react to complex traffic patterns that are difficult to model explicitly.

3.1 Fundamentals

The RL problem is formulated as a Markov Decision Process (MDP),

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P represents the traffic dynamics, and r is the reward.

At each time step, the agent observes the traffic state, selects an action, applies it to the controller, and receives a reward. In the current implementation, the state of each CAV i is defined as

$$x_i = (\mu_{v,i}, \sigma_{v,i}^2, \delta_{v,i}^{\mathrm{macro}}, v_i, s_i, \Delta v_i, a_{\ell(i)}, \alpha_i^{prev}),$$

where $\mu_{v,i}$ and $\sigma_{v,i}^2$ are the upstream mean velocity and velocity variance, $\delta_{v,i}^{\mathrm{macro}} = v_i - \mu_{v,i}$ is the speed mismatch with the local upstream flow, v_i is the ego speed, s_i is the spacing to the leader, Δv_i is the relative speed, $a_{\ell(i)}$ is the leader acceleration, and α_i^{prev} is the previously applied headway scaling factor.

This state combines *microscopic* information, such as spacing and relative velocity, with *mesoscopic* information, such as the average speed and speed variance of upstream traffic. This

is important because local measurements alone may not be enough to detect the development of traffic waves.

The action is a residual correction of the headway scaling:

$$\mathcal{A} = \{\Delta\alpha_i \in [-\Delta\alpha_{\max}, +\Delta\alpha_{\max}]\}.$$

The final headway scaling factor is then given by

$$\alpha_i(t) = \text{clip}\left(\alpha_i^{rule}(t) + \Delta\alpha_i(t)\right),$$

where $\alpha_i^{rule}(t)$ is the baseline headway scaling provided by the rule-based mesoscopic module. Therefore, RL only refines the baseline decision instead of replacing it. The learned policy can be written as

$$\pi : \mathcal{S} \to \mathcal{A}, \qquad \Delta\alpha_i(t) = \pi(x_i(t)).$$

3.2 Reward design

The reward is designed to stay aligned with the engineering goals of the project rather than with a purely numerical score:

$$r_t = r_{\text{sf}} + r_{\text{sp}} + r_{\text{ef}} + r_{\text{cf}} + r_{\text{ss}} + r_{\sigma^2}.$$

For the spacing-regulation term, the controller uses the desired gap

$$s_i^{\text{des}}[t] = d_0 + h_i'[t]\, v_i[t],$$

and the tracking error

$$e_i[t] = s_i[t] - s_i^{\text{des}}[t].$$

The different reward terms have a clear physical meaning:

- r_{sf}: safety term, penalizing unsafe spacing and too small time headway;
- r_{sp}: spacing regulation term, encouraging tracking of the desired spacing;
- r_{ef}: efficiency term, encouraging the vehicle speed to remain close to the upstream mean speed;
- r_{cf}: comfort term, penalizing high jerk;
- r_{ss}: string stability term, penalizing amplification of spacing errors along the platoon;
- r_{σ^2}: traffic smoothness term, penalizing large velocity variance in the traffic stream.

Taken together, these terms encourage a policy that is safe first, but also smooth and useful at traffic scale. A controller that improved one vehicle while making the overall flow worse would not be acceptable here.

3.3 PPO

Proximal Policy Optimization (PPO) is a policy-gradient RL algorithm that updates the policy in a stable and conservative way. Its main advantage is that it is simple to implement and usually robust during training. PPO avoids large policy updates by using a clipped objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right],$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)},$$

and $\hat{A}_t$ is the estimated advantage. In this project, PPO mainly serves as a strong baseline: it is simple to implement, reasonably stable, and easy to interpret when comparing learning behavior across runs.

3.4 SAC

Soft Actor-Critic (SAC) is another RL algorithm for continuous control. Unlike PPO, SAC is an off-policy method and includes an entropy term in its objective:

$$J(\pi) = \sum_t \mathbb{E}\left[r(s_t, a_t) + \beta \mathcal{H}(\pi(\cdot|s_t)) \right],$$

where $\mathcal{H}(\pi(\cdot|s_t))$ is the policy entropy and β controls the exploration level.

SAC is well suited to continuous-action problems and stochastic environments. Its replay buffer allows reuse of past experience, which can improve sample efficiency compared with on-policy methods such as PPO. This makes it a natural alternative here because the action is continuous and the mixed-traffic environment is noisy due to stochastic HDV behavior.

3.5 Role of RL in the framework

In our setup, RL does not produce the full acceleration command. It only adjusts the headway scaling around a rule-based mesoscopic baseline. This design choice is deliberate: it keeps the controller interpretable, limits unsafe departures from a known operating point, and makes it easier to judge whether learning is adding value beyond the handcrafted adaptation.

4 System Architecture

The proposed controller is organized as a layered architecture that combines a conventional vehicle-level controller with mesoscopic traffic awareness. Figure 1 summarizes the four layers: simulation, perception, adaptation, and control.

4.1 Overall structure

At each simulation step, the environment updates all vehicles, the perception layer summarizes the local and upstream traffic state, the adaptation layer converts those observations into a headway adjustment, and the control layer applies the resulting acceleration command. Separating these roles makes the framework easier to debug and makes it explicit what the RL block is allowed to change.

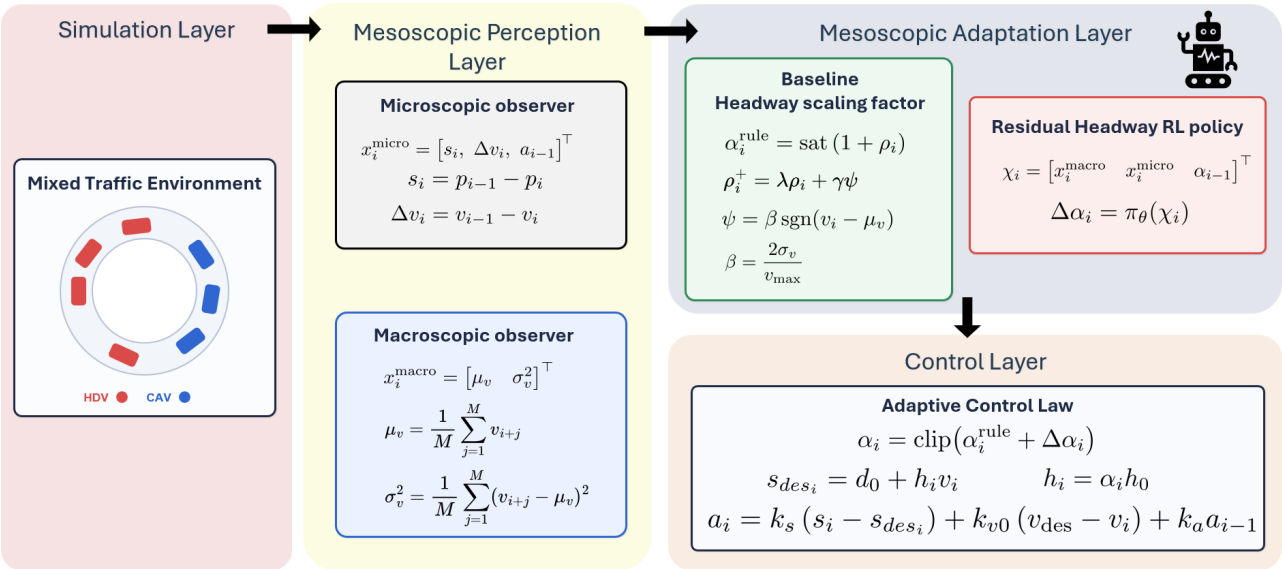

Figure 1: Overview of the proposed system architecture.

4.2 Simulation layer

The simulation layer models a single-lane ring road where HDVs and CAVs coexist. The ring-road setting removes open-boundary entry and exit effects, which makes the appearance of stop-and-go waves easier to observe and compare across methods. This layer provides the positions, velocities, spacing, and leader-acceleration signals used by the upper layers.

4.3 Mesoscopic perception layer

The perception layer extracts the information used by the controller. It contains two complementary observers.

Microscopic observer. The microscopic observer describes the local car-following situation of vehicle i. It uses the spacing, relative velocity, and ego speed:

$$x_i^{\text{micro}} = \begin{bmatrix} s_i & \Delta v_i & a_{i-1} \end{bmatrix}^\top,$$

with

$$s_i = p_{i-1} - p_i, \qquad \Delta v_i = v_{i-1} - v_i.$$

These variables describe the immediate interaction with the preceding vehicle and are the standard quantities used by longitudinal controllers.

Macroscopic observer. The second observer captures collective traffic behavior upstream of the CAV. Instead of looking only at the leader, it summarizes the traffic state over a perception horizon. The two main quantities are the upstream mean velocity and the upstream velocity variance:

$$x_i^{\text{macro}} = \begin{bmatrix} \mu_v & \sigma_v^2 \end{bmatrix}^\top,$$

with

$$\mu_v = \frac{1}{M}\sum_{j=1}^{M} v_{i+j}, \qquad \sigma_v^2 = \frac{1}{M}\sum_{j=1}^{M}(v_{i+j} - \mu_v)^2.$$

The mean velocity gives an estimate of the average upstream traffic speed, while the variance indicates how irregular or unstable the traffic stream is.

4.4 Mesoscopic adaptation layer

The adaptation layer transforms the observed traffic information into a headway modification for the CAV controller. It contains two parts: a baseline adaptive headway mechanism and a residual reinforcement learning policy.

Baseline adaptive headway. The baseline mechanism computes a rule-based headway scaling factor from the macroscopic traffic information. The update can be written as

$$\alpha_i^{\text{rule}} = \text{sat}(1 + \rho_i),$$

with the internal state

$$\rho_i^+ = \lambda \rho_i + \gamma \psi,$$

and the perception signal

$$\psi = \beta \, \text{sgn}(v_i - \mu_v), \qquad \beta = \frac{2\sigma_v}{v_{\max}}.$$

This mechanism has a simple interpretation. When upstream traffic becomes more variable, the controller increases the effective headway in order to react more cautiously. When traffic is smoother, the headway can be reduced. The filtering through ρ_i also avoids abrupt parameter changes.

Residual RL policy. On top of the baseline adaptation, a residual policy computes a correction term:

$$\chi_i = \begin{bmatrix} \mu_v & \sigma_v^2 & (v_i - \mu_v) & v_i & s_i & \Delta v_i & a_{i-1} & \alpha_i^{prev} \end{bmatrix}^\top, \qquad \Delta\alpha_i = \pi_\theta(\chi_i).$$

The role of this block is to refine the rule-based decision without changing the overall structure of the controller. In other words, the adaptive law remains physically interpretable, while the learning block adds flexibility when the traffic situation becomes more complex. The resulting policy input therefore combines mesoscopic information, local car-following variables, and the previously applied headway state in a single 8-dimensional vector.

4.5 Control layer

The control layer converts the adapted headway into a longitudinal acceleration command for the CAV. The final scaling factor is

$$\alpha_i = \text{clip}\big(\alpha_i^{\text{rule}} + \Delta\alpha_i\big),$$

and the corresponding time headway becomes

$$T_i = \alpha_i T_0.$$

This adapted headway is then used inside the microscopic CACC/CTH controller:

$$a_i = k_s \left(s_i - d_0 - T_i v_i\right) + k_v \Delta v_i + k_a a_{i-1} + k_{v0}(v_{\text{des}} - v_i).$$

The lower-level controller therefore remains a standard feedback law; the upper layers only adapt the time-headway parameter used inside it. This separation preserves a clear physical interpretation of the final command.

5 Experimental Results and Perspectives

5.1 Protocol and admissibility fix

The experiment targets the 50 % human-driver regime on a ring road with N=22 vehicles (L=300 m). We compare three methods: the rule-based mesoscopic adaptation alone (**No RL**), and two residual RL policies (**PPO**, **SAC**) trained on top of it. Each episode starts from random initial conditions, and a velocity perturbation ($\Delta v = -3$ m/s) is applied at t=12 s after a 10 s warm-up. The primary metric is the *string-stability amplification ratio*: the maximum downstream-to-source speed-error ratio, where a value below 1 indicates disturbance attenuation.

A critical protocol bug was found and fixed before this campaign. The random initialization computed minimum inter-vehicle spacing without accounting for vehicle length (ℓ=5 m), producing net bumper-to-bumper gaps as low as 0 m. Combined with a warm-up that also capped braking, earlier runs suffered collision-clamp events *before* the perturbation, invalidating string-stability. The fix added vehicle length to the spacing bound, clamped initial closing rates to be kinematically safe, and allowed unrestricted braking during warm-up. In the saved training and evaluation runs used for this report, the pre-perturbation clamp count is zero throughout.

5.2 Results

Both RL methods early-stopped quickly (PPO at episode 15, SAC at 21), with best checkpoints at episodes 4 and 6. In practice, most of the useful learning happened almost immediately. Given the limited residual authority ($|\Delta\alpha| \leq 0.1$), this is not surprising: the policy can refine the baseline, but it cannot move very far away from it. Zero collision clamps were observed throughout training, which is consistent with the corrected initialization.

Table 1 presents the evaluation. All three runs are fully admissible: zero collisions, zero clamps, perturbation correctly applied, and valid string-stability measurement.

Table 1: Evaluation at 50 % human ratio. All runs admissible. Bold = best per column.

Method	Amp. ratio (–)	Var_{100} ($\mathrm{m}^2/\mathrm{s}^2$)	Mean spd_{100} (m/s)	RMS acc (m/s^2)	Min gap (m)	Clamps
No RL	0.873	0.517	**4.408**	**0.399**	2.875	**0**
PPO	0.881	0.687	4.183	0.403	2.929	**0**
SAC	**0.873**	**0.482**	4.325	0.425	**2.957**	**0**

5.3 Interpretation

The mesoscopic baseline is already strong. The corrected no-RL run is already string stable, with an amplification ratio of 0.873. This matters because it shows the hand-designed architecture is doing real work on its own, rather than serving only as a scaffold for RL.

RL adds only a modest gain in this regime. SAC essentially matches the baseline on the primary metric while reducing late-regime speed variance by about 7 % and slightly increasing the minimum gap. PPO is a little worse than the baseline. With only ±10 % headway modulation, neither algorithm can move far from the rule-based policy.

SAC is the more convincing residual learner. Of the two RL methods, SAC is the only one that consistently improves the secondary metrics. That makes it the better candidate for future runs, even if the current gain is still small.

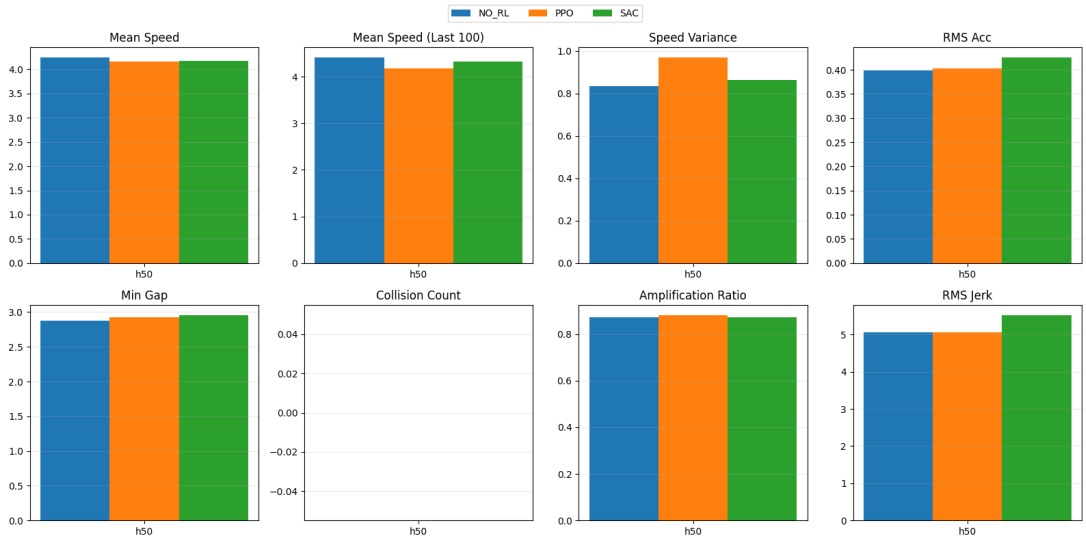

Figure 2: Comparison of the admissible h50 evaluation metrics. SAC gives the lowest late-regime speed variance, while the rule-based baseline remains competitive on speed and control effort.

The main result is methodological as much as numerical. The comparison is now admissible. That matters because earlier runs were contaminated by pre-perturbation clamps, so their string-stability scores could not be trusted.

5.4 Future work

The next priority is **multi-seed evaluation** (10–20 seeds) to separate genuine method differences from seed artefacts. After that, **increasing the RL authority** ($|\Delta\alpha|$ from 0.1 to 0.2–0.3) would give the policy more room to differentiate from the baseline. The campaign should also be extended to **25 % and 75 % penetration rates**, and training should move to the **GPU-vectorized backend** to enable longer runs with larger batch sizes.

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